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Head shape analysis of National Football League Players

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Abstract

Understanding the size and shape variations of the head is important to the design of football helmets used to mitigate the risk of head injury. Current guidelines for selecting helmets use only the circumference of the player's head to determine an appropriate size that may not offer the best fit and protection for every player. The goal of this study was to quantify key measurements of head shape to sufficiently characterize variability among players in the National Football League (NFL), focusing on the shape variation of the cranial region. Statistical shape analysis was performed on three-dimensional head scans of 87 contemporary NFL players to identify key features of head shape variability among the players' heads. The principal component analysis revealed two factors that explained 87% of the head shape variance: (1) height and length of the head in the sagittal plane and (2) prominent back of the head with an oval shape in the transverse plane versus prominent forehead with a round shape. Four head shape measurements (circumference, length, breadth, and height) were then defined and quantified to describe the first two principal components. This information can facilitate improvements in the design of football helmets.

Keywords

Football helmets, head shape, statistical shape analysis, principal component analysis, helmet fit

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Introduction

Football helmet technology has improved tremendously from the early designs of leather suspension helmets to modern helmets with polycarbonate shells utilizing different materials and structures to manage impact energy.[1] Current guidelines for selecting helmets use the player's head circumference as the primary determinant of the appropriate helmet size,[2–5] and inflatable air bladder systems and replaceable jaw pads with different thicknesses of comfort padding have been utilized in helmet models to allow for fit adjustment. Recent innovations include the development of a custom-fit helmet that personalizes the thickness of padding made of three-dimensionally (3D) printed lattice structures.[2] A custom-fit approach has the potential to improve the safety performance of helmets by optimizing fit and maximizing the use of available space for energy-absorbing materials.[6–8]

Biomechanical efforts regarding head shape and helmet performance have been mainly focused on gaining a better understanding of the effect of helmet fit on performance and concussion risk, although consensus has not been reached. While some studies suggest that properly fitted helmets may reduce concussion risk,[6–8] other studies have contended that a loosely fitted helmet, compared to a tightly fitted one, may lead to reduced head kinematics in some scenarios.[9] In addition to linking helmet fit and performance, Cobb et al. compared the shapes of two headforms (NOCSAE and Hybrid III) used for laboratory helmet testing and the study showed significant shape differences in several regions that translated to variations in the evaluation of helmet performance.[10] Cobb et al. also pointed out that those two headforms represent the median size of an adult male, which limits their applicability to the NFL population.

Statistical shape analysis (SSA) based on 3D head surface scans have been used extensively to understand

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head shape variability among the human population.[11–13] Ball et al.[13] studied head shape differences between Chinese and Caucasian populations using the SSA and observed a statistically significant difference in head shape between those two ethnic groups. Zhuang et al.[12] used the SSA to analyze head and face shape variations among United States civilian workers and observed differences by sex and ethnicity. In 2015, Lacko et al. applied principal component analysis (PCA) to 100 MRI scans of the human head and regressed the principal components with head shape measurements to develop a statistical shape model of the human head for improving the ergonomics of brain-computer interface headsets.[11] While these studies provide a framework for using the SSA to analyze variations in head shape across the general civilian population, there is little information available regarding head shape variations among football players.

The present study expands on these previous studies using the SSA and utilizes 3D head surface scans from a contemporary sample of players in the National Football League (NFL). SSA was performed to identify the main principal components of head shape variation among the scanned players, which were then used to develop key parameters for describing head shape variability. This study focused on the shape variation of the cranial region because it is the primary region of contact between the helmet and head, with direct helmet impacts being the primary source of concussion incidence in the NFL.[14–16]

Methods

Head shape measurements of professional football players were quantified through a stepwise process. First, 3D light scans of a sample of NFL players' heads were collected and processed for analysis. Next, head scan data were cleaned and aligned, and the SSA was used to identify the principal components (PCs) that explained the majority of the shape variability among players' heads. Relevant head shape measurements were then defined based on the main PCs. These measurements were tabulated for the sample of head scans and compared to those from the ANSUR II anthropometric survey, which consists of anthropometric measurements of U.S. Army soldiers, to determine whether the head shapes of the NFL player population differ from a wider range of the population.[17] The details of each of these steps are described below.

Head scans

A total of 129 players across three NFL teams representing various player-positions were scanned prior to the 2019–2020 NFL season. Player participation in the study was obtained through written consent based on a protocol that was approved by the NFL and NFLPA Medical Research Approval Process[18] and Institutional Board at the (blinded for editorial review) School of Medicine. Head scans were collected using a 3D structured light scanner with a spatial resolution of 0.5 mm (Artec Eva, Artec 3D, Santa Clara, CA). Players were scanned in an upright seated posture wearing a snug nylon cap to improve scan quality around the hair. Raw de-identified head scans were processed and exported to 3D mesh models of each player's head (Artec Studio 13, Artec 3D, Santa Clara, CA). A subset of 87 head scans was selected for the SSA; scans from players with longer hairstyles (e.g., dreadlocks and ponytails) were excluded. The average height (188.5 ± 7.0 cm) and weight (110.3 ± 21.7 kg) of the samples were comparable (statistically non-significant, $p > 0.05$) to the population of NFL players ($N = 11{,}790$) during the 2015–2018 seasons (height: 188.2 ± 6.7 cm, weight: 111.0 ± 21.0 kg), which was obtained from the NFL Game Statistics and Information System (GSIS) database.[15,19] The distribution of the height and weight between the NFL player population and the samples are compared in supplementary material (Supplemental Figure A1).

Statistical shape analysis (SSA)

The 3D head scans were then processed and aligned for the SSA (Figure 1). First, five anatomical landmarks (the nasion, left/right lower orbits, and left/right tragions) were manually selected to define the local head anatomical coordinate system. The head center-of-gravity (CG) was defined by locating the origin 92 mm posterior and 3 mm inferior to the nasion and the horizontal (xy-plane) was established parallel to Frankfurt plane based on the lower orbits and the tragions.[20] Head scans were then registered based on the origin and orientation of the local head coordinate system for each scan.

Next, the head scans were imported by Meshmixer (v3.5, Autodesk, San Rafael, CA) and prepared for the SSA. For each scan, faces were resized to have a nominal value of 3.5 mm, and irregular surfaces were smoothed, as necessary.[21] A set of pseudo-landmarks on the scalp of the heads was obtained from each cleaned head scan to ensure the correspondence between head scans for the SSA (Figure 1). The scans were sectioned at the scalp plane, defined as the transverse plane pitched (rotated in Y-axis) by 15° with respect to the head coordinate system, and the geometry inferior to the plane was defined as the scalp. The geometries of the scalp were mapped to the spherical coordinate system and the pseudo-landmarks were identified for every 5° in azimuth ($-\pi$ to $+\pi$) and elevation ($-\pi$ to 0) ($N = 1278$).

To ensure that the selected pseudo-landmarks sufficiently captured the characteristics of the scalp shape and to visualize the mode of PCs, 3D head shapes were reconstructed, implementing a thin-plate spline with a radial basis function.[22] This morphing technique transforms a template head shape (a randomly selected sample among 87 samples) to the target shape using the

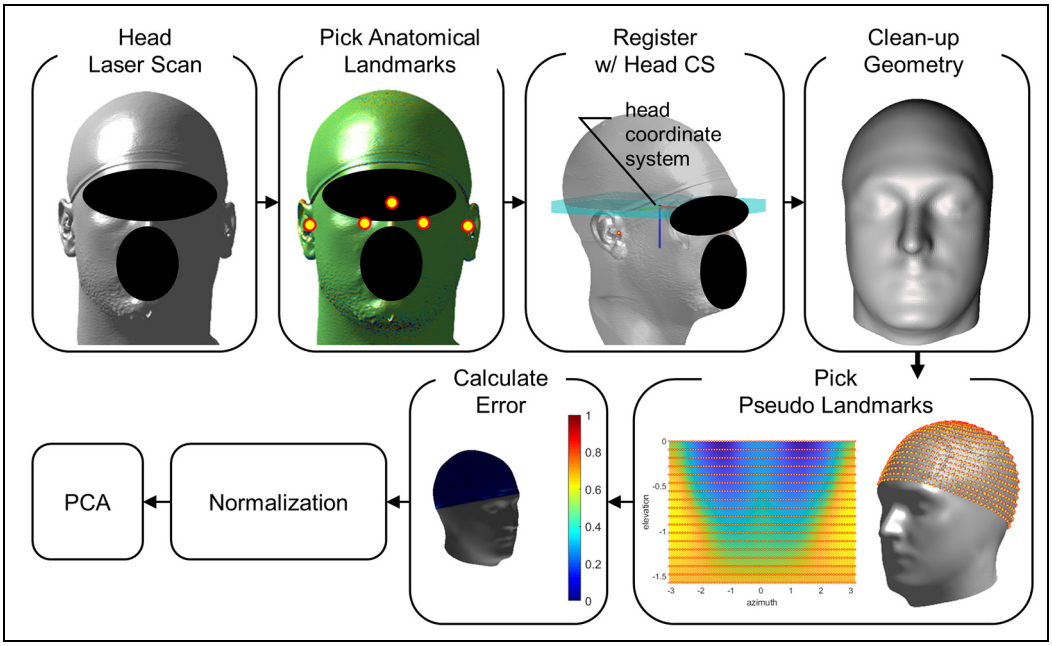

Figure 1. Schematic diagram of the overall procedure.

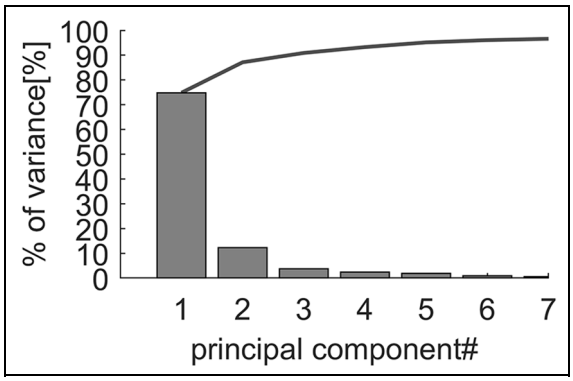

Figure 2. Scree plot of percentage of the variance of principal components.

1278 pseudo-landmarks selected for the SSA. The error from this morphing procedure, defined as the average of the distance between the vertices of morphed surface to the target surface (Supplemental Figure A2), was 0.03 ± 0.02 mm for the scalp across 87 samples (Figure 1). The set of pseudo-landmarks was then normalized by the average circumference of cross-sectional contour on the scalp plane to exclude size from the shape factors from the SSA.[23] A correlation-based PCA was performed using the "PCA" function of the statistical toolbox in MATLAB (R2019a, The MathWorks Inc., Natick, MA). The correlation matrix (1278×87) was constructed with a normalized radius of 1278 pseudo-landmarks from 87 head scans.

Head shape measurements

A scree plot was used to analyze a relative contribution of each PCs explaining the shape variability and to select the key PCs. Head shape measurements were defined to reflect the key PCs from the SSA and then quantified for each of the 87 head scans. To compare the specificity in head shapes of the NFL player population to a wider range of the human population, distributions of the head shape measurements from the NFL players' sample analyzed in this study were compared to those from the publicly available ANSUR II database, which consists of anthropometric measurements of U.S. Army soldiers from 2010 to 2012.[17] The head shape measurements of 4082 male soldiers were compiled from the database in this study (Table 2).

Results

The results of the SSA showed that the first two PCs explained 87% of the shape variation in the players' heads (Figure 2) and the eigenvalues of the first seven PCs are presented in Supplemental Material (Supplemental Table A1). The modes of these PCs can be illustrated by the cross-sectional geometry of head in the transverse and sagittal planes (Figure 3). The first PC, which explained 74.8% of shape variation, can be described as a tall or short head shape, which is mainly related to the height and length of the head in the sagittal plane. The second PC explained 12.3% of shape variation and can be delineated as a prominent back of the head with an oval shape in the transverse plane versus prominent forehead with a round shape.

The SSA showed that variations in the first two PCs could be described using four commonly used head shape measurements: circumference, length, breadth, and height. The correlation plots of four head shape measurements and PC scores for the first two PCs are provided in the Supplemental Material (Supplemental Figure A3). To capture the tall and short head shape,

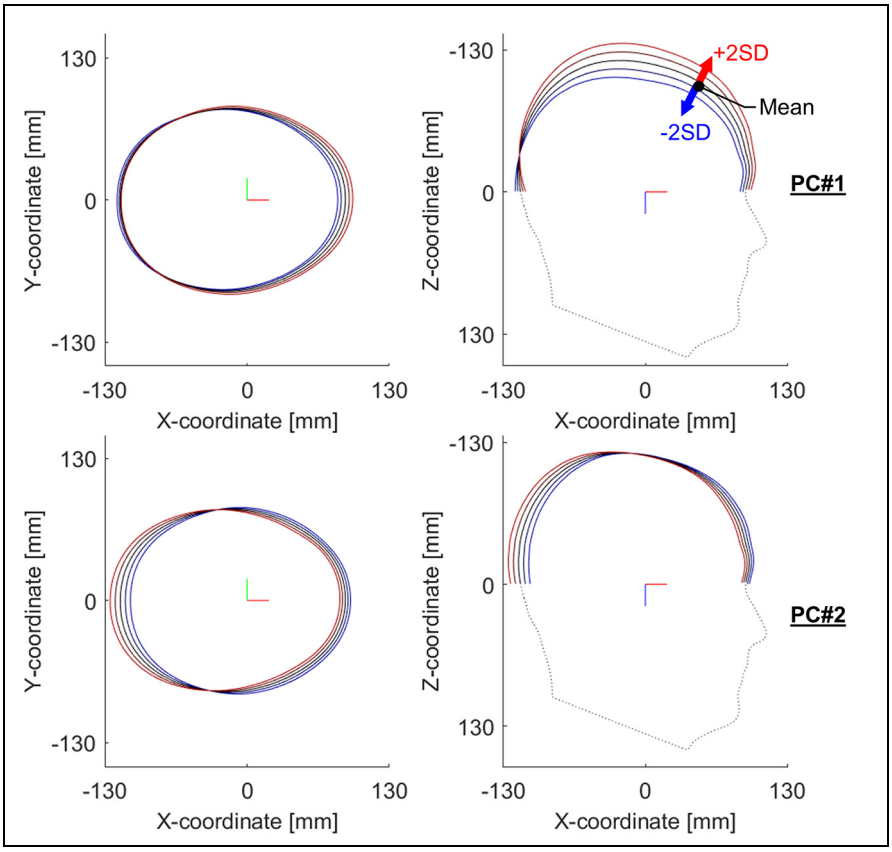

Figure3. Principal components of the head: cross-sectional geometries of the head on the helmet fit plane (left, Table 1) and the sagittal plane (right).

Table 1. Definitions of head shape measurements.

	Definition
Circumference (*C*)	Circumference of the head measured from the cross-sectional contour on the helmet fit plane. The helmet fit plane was defined as the transverse plane pitched to position one inch above the glabella and occipital landmarks.
Length (*L*)	The perpendicular (*x*-axis) distance between the glabella and occipital landmarks of cross-sectional geometry in the sagittal plane.
Breadth (*B*)	The maximum horizontal (*y*-axis) distance of the upper skull part of the head in 3D shape.
Height (*H*)	The maximum superior vertical distance (z-axis) between the average z-coordinate of tragions and the cross-sectional contour of the head on the sagittal plane.

the height and length of the head should be measured. Besides, the cross-section of the head shape in the transverse plane, whether it is oval or round, can be delineated by an aspect ratio (AR) of the cross-sectional shape, which can be defined as the ratio of the breadth to length. Because circumference was used to normalize the data to focus on head shape, it was included as one of the four measurements to describe head size. These four measurements were defined to be consistent with those used by the ANSUR II dataset (Table 1 and Figure 4).[17] When compared to the ANSUR II dataset, the circumference, length, and breadth measurements for NFL players were larger on average (statistically significant, $p < 0.05$) (Table 2), while correlations among various head shape parameters were similar between the groups (Figure 5).

Discussion

This study analyzed the head shape variability among NFL players using the SSA and quantified four head shape measurements intended to describe the PCs, which reflect the key shape attributes of NFL players' heads. The results of the SSA suggest that the head shape of this sample of NFL players can be categorized in two separate ways: (1) by height and length of the head in the sagittal plane (tall vs short) and (2) by the length and breadth of the head (prominent back with an oval shape in the transverse plane versus prominent forehead with a round shape (Figure 3). The first PC observed in this study is consistent with Ball et al.,[13]

Table 2. Head shape measurements.

	NFL [mm]	ANSUR II [mm]	*p*-Value +	Effect size ++
C	592.3 ± 17.2	574.4 ± 16.0	< 0.05	1.1
L	212.2 ± 7.1	199.5 ± 7.0	< 0.05	1.8
B	168.2 ± 6.3	154.3 ± 5.5	< 0.05	2.4
H	152.8 ± 6.0	131.1 ± 6.2	< 0.05	0.4

C: circumference; *L*: length; *B*: breadth; *H*: height.
+ Student *t*-test.
++ Hedges'*g*.

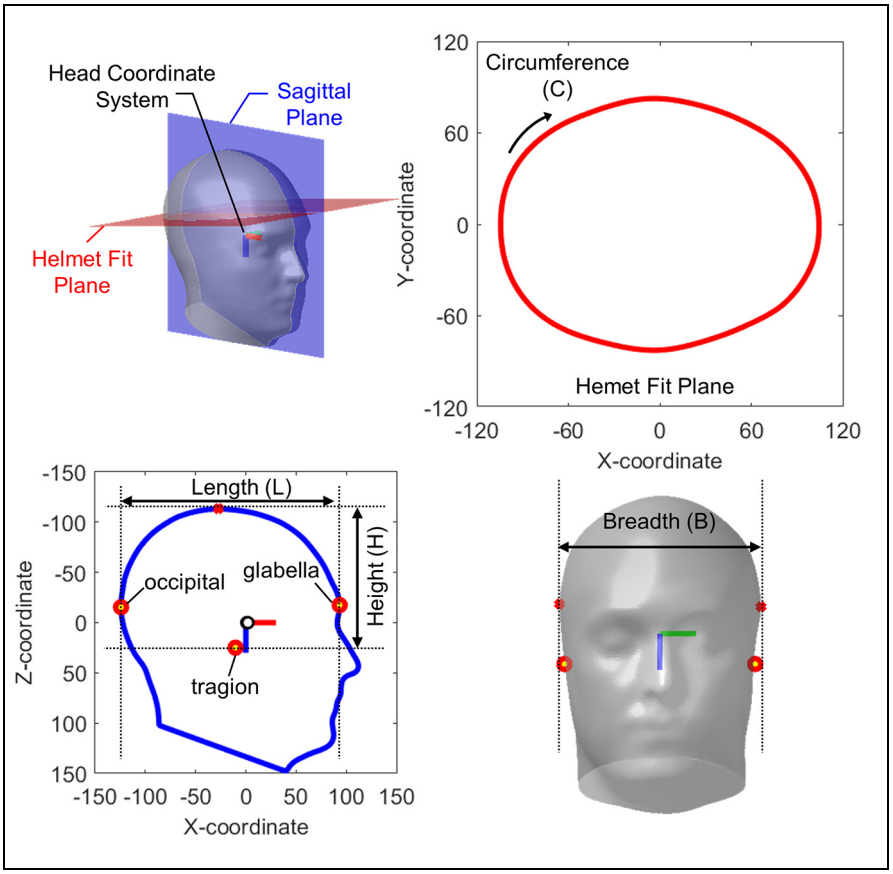

Figure 4. Visual representation of anatomical landmarks and head shape parameters in the head local coordinate system.

who determined in an analysis of head shapes of Caucasian and Chinese populations that the height of the head was the second PC following the size of the head. Lacko et al.[11] also identified that the length and height were key PCs to characterize the shape of the human head (100 adults from the Western population). Note that the first PC in the analysis by Ball et al.[13] and Lacko et al. was the size, which was removed from the current data set prior to the SSA by normalization. Ball et al. also observed variation in oval versus round shapes of the head (the second PC of the current study) and attributed those differences between Caucasian and Chinese populations.

When designing the components of non-custom helmets, information on the head shape attributes obtained from this study could prove useful in optimizing and narrowing down the number of shell/padding combinations necessary to provide improved fit and performance across the general population of professional football players. Current helmet models rely on one baseline design for a given helmet model, offering different sizing options based on the player's head circumference. Different helmet sizes typically are achieved in two ways: (1) scaling the three-dimensional shell size or (2) scaling the size of the internal padding components while using the same shell. Players may choose a different helmet size from the size recommended by a manufacturer due to their head shape attributes and how well the shape attributes match with a baseline helmet design. For example, a baseline design of one helmet was optimized to a head with a round head shape. In this case, a player who has an oval head

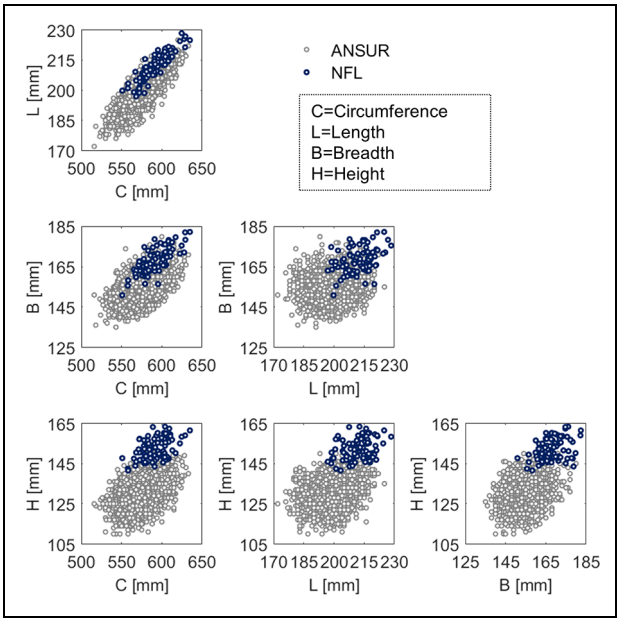

Figure 5. Comparison of head shape measurements between ANSUR II database and NFL samples.

shape would choose a larger size helmet since the helmet may fit too tightly because of the player's head length, while a player with a round head shape may be comfortable with the helmet size recommended from the manufacturer. Alternatively, a player with an oval head shape may choose a different helmet model that offers a baseline design closer to an oval head shape. Utilizing the size and shape parameters described in this study would allow helmet manufacturers to categorize helmet designs in several different standardized head shapes and design helmets in a way that best optimizes the use of space within a given helmet model for target player populations. Also, manufacturers could offer more specific size and fitting guidelines based on more shape measurements utilizing the results of this study.

Given the annual NFL and NFL Players Association helmet testing initiatives,[24] another possible application of this work relates to the development of NFL player-specific headform for laboratory helmet tests. Since the NFL player population is not representative of the overall male population, current headforms used for laboratory testing of football helmets do not accurately represent the size, shape, and mass of the heads of NFL players.[10,25–27] The 50th percentile Hybrid III headform, which were developed to target the median male size of the American population,[10] underestimates the size and physical characteristics (mass and moment of inertia) of the head of NFL players.

While a custom-helmet approach aims to improve helmet performance by optimizing the use of available space between the head and helmet, it does not necessarily imply that the entire helmet design has been optimized for individual players' heads. Even with personalized padding thickness for an individual player, the same helmet shell design may not provide equivalent protection for all players having different head shapes. For example, a given helmet shell design would affect the available space for padding, which could potentially reduce the effectiveness of a custom helmet in certain impact locations. Although fully optimizing both the shell and padding components for individual players could be costly from a manufacturing standpoint given current technologies, a future initiative of custom-helmet designs may need to account for head shape variability throughout the entire design process, not limiting customization to just padding thickness.

This study is subject to some limitations. Scans from a sample of approximately 5% of current NFL players were used for analyses. An increased number of head scans from NFL players would potentially enhance the range of application of the current study. While the sampled NFL player population used in this study had comparable weights and heights to those of the entire player population, the number of scans collected was insufficient to analyze the difference of the head shape variations between player positions. Having adequate data on head shape variations occurring within player positions may help in the development of position-specific helmets. Second, this study excluded players with certain hairstyles to minimize the effect of hair on the analysis. Reed and Corner[28] utilized a head-borne device with probes around a scalp to contact the head to characterize cranial surface contour. While obtaining a cranium shape of the head supplements the outline surfaces from head scans, complete removal of the hair for the shape analysis is also not ideal, since the helmet is fit with hair. Future research is needed to address how hair affects the helmet fit and safety performance.

Understanding the attributes of shape variability among NFL players is important to improve the design and selection of helmets for comfort and injury mitigation. This study identifies the key head shape measurements (circumference, length, breadth, and height) with SSA utilizing 3D head surface scans from a contemporary sample of players and quantifies them. This information can facilitate improvements in the design of football helmets.

Declaration of conflicting interests

The author(s) declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article: Author Jeff Crandall has ownership interest in a company (Biomechanics Consulting and Research, LC) that receives consulting fees from the Football Research, Inc. including for the conduct of the study submitted for publication. Authors Park, Gabler, Bailey, Dau, and Sherwood are employees of Biomechanics Consulting and Research, LC. Authors Arbogast has received consulting fees from the National Football League Players Association.

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Supplemental material

Supplemental material for this article is available online.

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